

# Session 2

## Case Study: The UK Data Landscape

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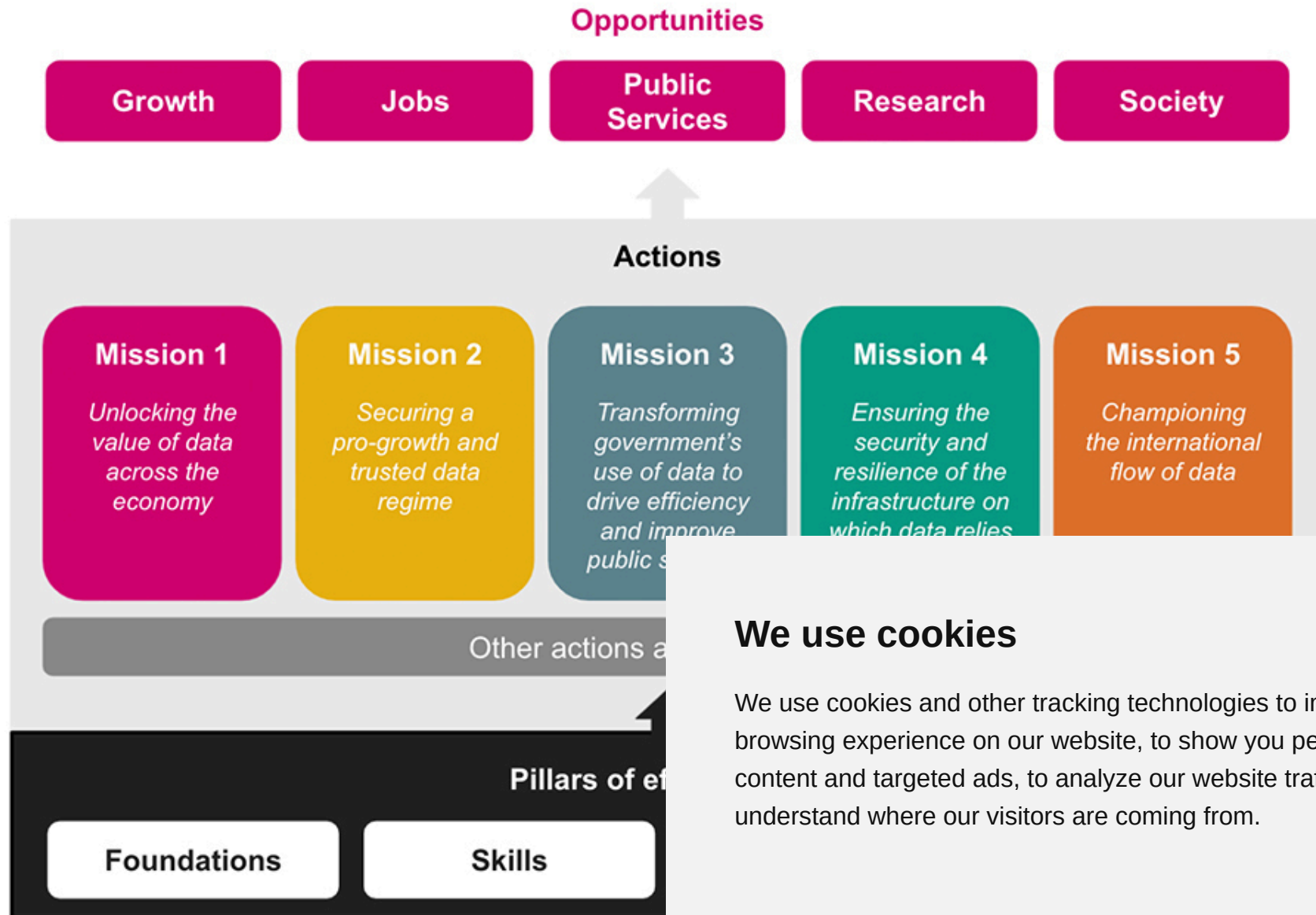


# National Data Strategy

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# National Data Strategy Framework



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# Core pillars of the UK National Data Strategy

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# Data foundations

- Data reaches its full potential only when it is high-quality, fit for purpose, and recorded in standardised formats.
- It must be stored on modern, future-proof systems that ensure it is easy to find, access, combine, and reuse.
- By raising data quality, we can use it more effectively and achieve stronger insights and improved outcomes.

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# Data skills

- Making the most of data requires a strong pool of people with data skills.
- This involves providing the right training and knowledge through the education system.
- It also means ensuring individuals can keep developing and updating their data skills throughout their lives.

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# Data availability

- Data has the greatest impact when it is easy to access, move, and reuse.
- Achieving this involves encouraging better coordination and sharing of high-quality data among organisations in the public, private, and third sectors.
- It also requires ensuring strong and appropriate protections for the secure international fl

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# Responsible data

- As we expand the use of data, it must be managed in a responsible manner.
- Data use should always be lawful, secure, fair, ethical, sustainable, and accountable.
- We must also ensure that these safeguards continue to enable innovation and research.

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# Five priority areas of action

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# Unlocking the value of data across the economy

- Full potential of data is not being realised because important information is not reaching the people and places that need it.
- The initial focus is on creating the right conditions for data to be usable, accessible, and widely available throughout the economy.
- Ensure strong protection for individuals' data rights and for the intellectual property of private enterprises.
- Careful, evidence-based approach to design a clear policy framework that identifies the government interventions needed to achieve

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# Securing a pro-growth and trusted data regime

- Data revolution to benefit businesses of all sizes.
- Goal of creating a data regime that is easy for companies to work with and supports responsible, secure data use.
- Reducing unnecessary regulatory uncertainty or risk to help innovators drive economic growth.
- The public feels confident and empowered in how their data is used, including personal data.
- Encourage competition and innovation with high standards.
- Build trust without placing unnecessary barriers on businesses.

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# Transforming government's use of data to drive efficiency and improve public services

- Significant untapped potential in how government and public services use and share data to support people.
- Improve how information is managed, used, and shared across government.
- Requires a coordinated, whole-government approach that aligns around best practices and common standards for deriving value and insight from data
- Building a secure, connected, and interoperable data ecosystem with strong safeguards.
- Developing strong data skills and leadership to unlock data's full potential.

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# Ensuring the security and resilience of the infrastructure on which data relies

- Systems that support data must be safe and secure.
- Data use infrastructure is a critical national asset and should be protected from security threats and other risks, including service disruptions.
- Although managing these risks is partly a commercial responsibility, the government must ensure that data and its supporting infrastructure are resilient to both existing and emerging threats.
- Strengthening this resilience is essential for the economy.

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# Championing the international flow of data

- Cross-border data flows drive global business operations, supply chains, and trade, supporting economic growth worldwide.
- Data movement also provides social benefits, such as enabling salary payments and helping people stay connected across long distances.
- Sharing health data can advance vital scientific research and support coordinated international responses to global health emergencies.
- Promote strong domestic data practices and work with international partners to avoid unnecessary barriers created by national b

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# The data opportunity

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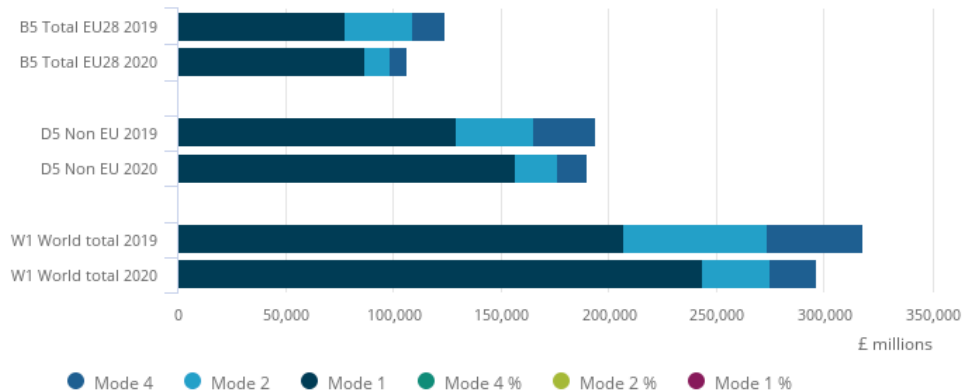
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# Boosting productivity and trade - digitally-delivered services

## Exports

Figure 2: In 2020, 82.1% of UK service exports were conducted remotely via Mode 1, an increase from 65.2% in 2019

UK services exports by value of mode of supply, 2020

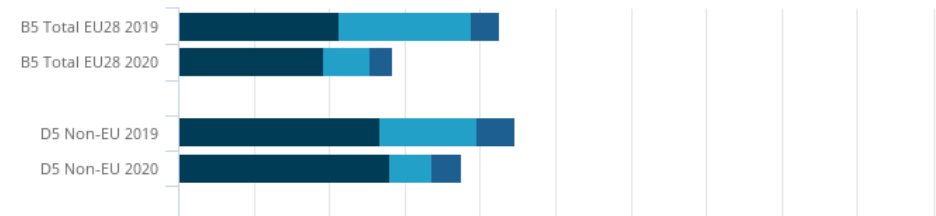


Source: Office for National Statistics - UK Trade in Services by Mode of Supply

## Imports

Figure 4: 71.8% of UK service imports were conducted remotely via Mode 1 in 2020, up from 54.9% in 2019

UK services imports by value of mode of supply, 2020



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# Boosting productivity and trade - Transport for London open data

Publishing open data creates a virtuous circle that benefits those using and delivering transport networks in the Capital

Transport Open data that can be freely used, re-used and redistributed by anyone can support operational service improvements, the development of new customer facing products and services, increase transparency and innovation and challenge existing ways of working.



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# Boosting productivity and trade - UK government inputs

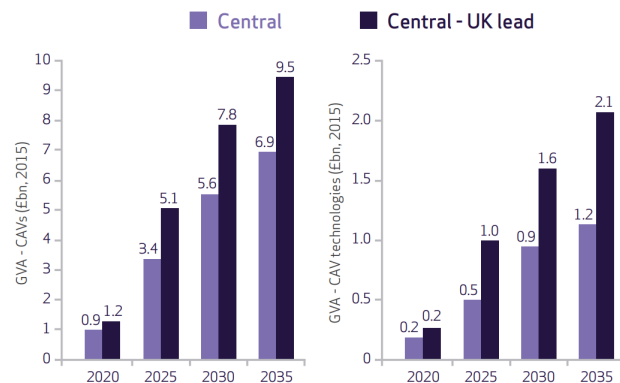
- It has committed to raising research and development investment, majority of which is data-intensive, to 2.4% of GDP by 2027.
- New institutions have been created, including the [Centre for Data Ethics and Innovation \(CDEI\)](#) and the [Alan Turing Institute](#).
- The government has introduced new data science and AI conversion courses.
- It has undertaken pioneering work on “*data trusts*,” a new framework for data sharing.

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# Supporting new businesses and jobs - digital economy and transport sector

## Gross value added (GVA)



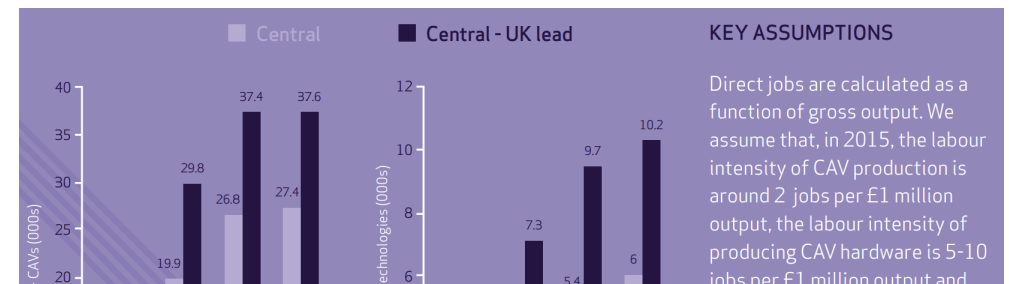
### KEY ASSUMPTIONS

GVA is calculated as gross output net of intermediate consumption, based on the UK Supply and Use tables.

**Central case:** Gross output in the CAV sector reaches £6.9bn by 2035 and GVA in CAV technologies reaches £1.2bn in the same year.

**Central - UK lead:** Gross output in the CAV sector reaches £9.5bn by 2035 and gross output in CAV technologies reaches £2.1bn in the same year.

## Jobs



### KEY ASSUMPTIONS

Direct jobs are calculated as a function of gross output. We assume that, in 2015, the labour intensity of CAV production is around 2 jobs per £1 million output, the labour intensity of producing CAV hardware is 5-10 jobs per £1 million output and

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# Increasing the speed, efficiency and scope of scientific research - NHS and clinical trials

- data-driven clinical trial to tackle coronavirus - the [RECOVERY](#) trial.
- using NHS electronic health records to identify and enable individuals to participate from across the UK.
- clinical trials normally take months to set up but RECOVERY was set up in a matter of days.
- 12,000 patients from 176 NHS hospital organisations being tracked and analysed.
- Within 100 days, the trial identified the worst treatment to save lives – *Dexamethasone*.
- RECOVERY was supported by [NHS DigiTrial](#).

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# Driving better delivery of policy and public services - Data First

- **Data First** is a major data-linking programme run by the [Ministry of Justice \(MoJ\)](#) and funded by [UK Research and Innovation](#).
- It provides secure access to anonymised, linked administrative datasets from the MoJ and its agencies, with the ability to connect these to data from other departments (e.g., the Department for Education).
- The programme helps researchers analyse how people interact with the justice system over time and identify factors linked to frequent court use.
- It enables deeper insight into how individual backgrounds shape their justice system experience.
- These insights will support more effective, services, and reduced costs.
- Data is accessed through the [ONS Secure Research](#) of security and data protection under the [Freedom of Information Act](#).

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# Creating a fairer society for all - ONS domestic abuse statistics tool

- The ONS domestic abuse publication combines data from multiple sources—including surveys, police records, government bodies, and support services—to provide a more complete national and local picture of domestic abuse.
- Integrating these data sources helps improve understanding of victims' experiences and assess how the criminal justice system responds to perpetrators.
- An accompanying interactive data tool allows users to explore and compare statistics by police force area, helping inform the questions and priorities for agencies addressing domestic abuse.
- These statistics also contribute to tracking Sustainable Development Goals under the

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